Tech Bite #5: The Math, Made Small

The biggest group on the exam -- and the most feared -- reduced to a handful of simple rules: the metric ladder, decibels, Ohm's Law, and power.

Groups T5B + T5C + T5D · 39 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

The Big Picture

This Bite has more questions than any other in the program – 39 – and it is the one beginners dread. Here is the secret: those 39 questions hide just a few small rules. Learn the rules and the whole group falls. Nothing here is harder than grade-school arithmetic, and the diagrams below do most of the remembering for you.

Rule 1: The Metric Ladder

Radio deals in very large and very small numbers, so it uses **prefixes**. Each step up or down the ladder is a factor of **1000**. To convert, you simply move the decimal point.

So 1.5 amperes is 1500 milliamperes; 1,500,000 hertz is 1500 kilohertz; one kilovolt is 1000 volts. About **ten** exam questions are nothing but this one ladder. (T5B01-T5B08, T5B12, T5B13)

giga (G)	x 1,000,000,000
mega (M)	x 1,000,000
kilo (k)	x 1,000
the basic unit	
milli (m)	a thousandth
micro	a millionth
pico (p)	a trillionth

The metric ladder. Each step is a factor of 1000. "kilo/mega/giga" go up; "milli/micro/pico" go down.

Rule 2: Decibels – the 3, 6, 10 Rule

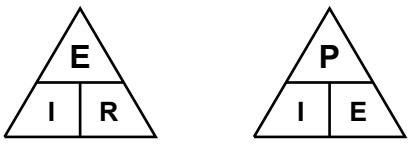
A **decibel** (dB) measures a change in power. Memorize three values and you can answer every decibel question:

Change	What it means	Q
+3 dB	Power **doubled** (x2). Example: 5 W up to 10 W is +3 dB.	T5B09
+/- 6 dB	Power **x4** or **1/4**. Example: 12 W down to 3 W is −6 dB.	T5B10
+10 dB	Power **x10**. Example: 20 W up to 200 W is +10 dB.	T5B11

Rules 3 and 4: Ohm's Law and Power

Ohm's Law links voltage (E), current (I), and resistance (R). The triangle does the work: cover the quantity you want, and the other two show the formula – **E = I x R**, **I = E / R**, **R = E / I**. (T5D01-T5D03) Every calculation question in group T5D is just plugging numbers in. A circuit with 90 volts and 3 amperes has a resistance of 90 / 3 = 30 ohms. (T5D04)

Power works the same way: **P = I x E**, with its own triangle. So 13.8 volts at 10 amperes delivers 13.8 x 10 = 138 watts. (T5C08, T5C09)



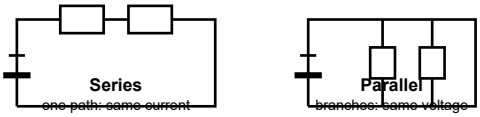
Ohm's Law: E = I x R **Power: P = I x E**

The two triangles. Cover the quantity you want; the triangle shows whether to multiply or divide.

A Few Definitions, and Series vs Parallel

Group T5C also has plain vocabulary. **Capacitance** is the ability to store energy in an **electric field**; its unit is the **farad**. (T5C01, T5C02) **Inductance** stores energy in a **magnetic field**; its unit is the **henry**. (T5C03, T5C04) **Impedance** is the opposition to **AC** current flow; its unit is the **ohm**. (T5C05, T5C12) The abbreviations are written **kHz** and **MHz**. (T5C06, T5C07)

Components can be wired two ways. In a **series** circuit there is one path, so the **current** is the same through every component. (T5D13) In a **parallel** circuit the components are branches, so the **voltage** is the same across each one. (T5D14)



Series: one path, current is shared. Parallel: branches, voltage is shared.

Mental Cheat Sheet

- Metric ladder: each step is **x1000**. kilo/mega/giga up; milli/micro/pico down.
- Decibels: **+3 dB = x2** power, **+6 dB = x4**, **+10 dB = x10**.
- Ohm's Law: **E = I x R**, **I = E / R**, **R = E / I**.
- Power: **P = I x E**, **I = P / E**, **E = P / I**.
- Capacitance stores energy in an **electric** field; unit **farad**.
- Inductance stores energy in a **magnetic** field; unit **henry**.

Common Traps

- 1.5 A = **1500** mA – "milli" is thousandths, so amps-to-milliamps multiplies by 1000.
- **3 dB is x2** power, not x3.
- Given volts and amps, asked for resistance? **Divide**: R = E / I.
- Capacitance stores in an **electric** field; inductance in a **magnetic** field – do not swap them.

- Impedance: opposition to AC; unit **ohm**.
- **Series = same current; parallel = same voltage.**

- **Series shares current; parallel shares voltage** – not the reverse.
- All radio frequencies and units use the same ladder – no special cases.

Takeaway: 39 questions, a handful of rules. The **metric ladder** (x1000 per step) handles the conversions. The **3/6/10 rule** handles decibels. The **Ohm's Law** and **Power** triangles – E = I x R and P = I x E – handle every calculation. Add three definitions (farad, henry, ohm) and the series/parallel rule, and the part of the exam everyone fears is done.

Bite 5 Question Bank (T5B + T5C + T5D)

T5B01. How many milliamperes is 1.5 amperes?
A. 0.0000015 milliamperes
B. 0.0015 milliamperes
C. 1500 milliamperes
D. 1,500,000 milliamperes

T5B02. Which is equal to 1,500,000 hertz?
A. 1500 kHz
B. 1500 MHz
C. 15 GHz
D. 150 kHz

T5B03. Which is equal to one kilovolt?
A. One one-thousandth of a volt
B. One hundred volts
C. One thousand volts
D. One million volts

T5B04. Which is equal to one microvolt?
A. One one-millionth of a volt
B. One million volts
C. One thousand kilovolts
D. One one-thousandth of a volt

T5B05. Which is equal to 500 milliwatts?
A. 5 watts
B. 0.5 watts
C. 500,000 watts
D. 500,000,000 watts

T5B06. Which is equal to 3000 milliamperes?
A. 0.003 amperes
B. 0.3 amperes
C. 3,000,000 amperes
D. 3 amperes

T5B07. Which is equal to 3.525 MHz?
A. 0.003525 kHz
B. 35.25 kHz
C. 3525 kHz
D. 3,525,000 kHz

T5B08. Which is equal to 1,000,000 picofarads?
A. 0.001 microfarads
B. 1 microfarad
C. 1000 microfarads
D. 1,000,000,000 microfarads

T5B09. Which decibel value most closely represents a power increase from 5 watts to 10 watts?
A. 2 dB
B. 3 dB
C. 5 dB
D. 10 dB

T5B10. Which decibel value most closely represents a power decrease from 12 watts to 3 watts?
A. −1 dB
B. −3 dB
C. −6 dB
D. −9 dB

T5B11. Which decibel value represents a power increase from 20 watts to 200 watts?
A. 10 dB
B. 12 dB
C. 18 dB
D. 28 dB

T5B12. Which is equal to 28400 kHz?
A. 28.400 kHz
B. 2.800 MHz
C. 284.00 MHz
D. 28.400 MHz

T5B13. Which is equal to 2425 MHz?
A. 0.002425 GHz
B. 24.25 GHz
C. 2.425 GHz
D. 242.5 GHz

T5C01. What describes the ability to store energy in an electric field?
A. Inductance
B. Resistance
C. Frequency
D. Capacitance

T5C02. What is the unit of capacitance?
A. Farad
B. Ohm
C. Volt
D. Henry

T5C03. What describes the ability to store energy in a magnetic field?
A. Admittance
B. Capacitance
C. Resistance
D. Inductance

T5C04. What is the unit of inductance?
A. Coulomb
B. Farad
C. Henry
D. Ohm

T5C05. What is the unit of impedance?
A. Volt
B. Ampere
C. Coulomb
D. Ohm

T5C06. What is the abbreviation for kilohertz?
A. kHz
B. khz
C. KHz
D. KHZ

T5C07. What is the abbreviation for megahertz?
A. mHz
B. mHZ
C. Mhz
D. MHz

T5C08. What formula is used to calculate electrical power (P) in a DC circuit?
A. P = I x E
B. P = E / I
C. P = I2 x E
D. P = I / E

T5C09. How much power is delivered by a voltage of 13.8 volts DC and a current of 10 amperes?
A. 138 watts
B. 0.7 watts
C. 23.8 watts
D. 3.8 watts

T5C10. How much power is delivered by a voltage of 12 volts DC and a current of 2.5 amperes?
A. 4.8 watts
B. 30 watts
C. 14.5 watts
D. 0.208 watts

T5C11. How much current is required to deliver 120 watts at a voltage of 12 volts DC?
A. 0.1 amperes
B. 10 amperes
C. 12 amperes
D. 132 amperes

T5C12. What is impedance?
A. The opposition to AC current flow
B. The inverse of resistance
C. The inverse of reactance
D. The power handling capability of a component

T5D01. What formula is used to calculate current in a circuit?
A. I = E x R
B. I = E / R
C. I = E2 x R
D. I = E2 / R

T5D02. What formula is used to calculate voltage in a circuit?
A. E = I x R
B. E = I / R
C. E = I2 x R
D. E = I2 / R

T5D03. What formula is used to calculate resistance in a circuit?
A. R = E x I
B. R = E / I
C. R = E + I
D. R = E - I

T5D04. What is the resistance of a circuit in which a current of 3 amperes flows when connected to 90 volts?
A. 3 ohms
B. 30 ohms
C. 1/30 ohm
D. 270 ohms

T5D05. What is the resistance of a circuit for which the applied voltage is 12 volts and the current flow is 1.5 amperes?
A. 18 ohms
B. 0.125 ohms
C. 8 ohms
D. 13.5 ohms

T5D06. What is the resistance of a circuit that draws 4 amperes from a 12-volt source?
A. 3 ohms
B. 16 ohms
C. 48 ohms
D. 8 ohms

T5D07. What is the current in a circuit with an applied voltage of 120 volts and a resistance of 80 ohms?
A. 9600 amperes
B. 200 amperes
C. 0.667 amperes
D. 1.5 amperes

T5D08. What is the current through a 100-ohm resistor connected across 200 volts?
A. 20,000 amperes
B. 0.5 amperes
C. 2 amperes
D. 100 amperes

T5D09. What is the current through a 24-ohm resistor connected across 240 volts?
A. 2400 amperes
B. 0.1 amperes
C. 10 amperes
D. 5760 amperes

T5D10. What is the voltage across a 2-ohm resistor if a current of 0.5 amperes flows through it?
A. 1 volt
B. 0.25 volts
C. 2.5 volts
D. 1.5 volts

T5D11. What is the voltage across a 10-ohm resistor if a current of 1 ampere flows through it?
A. 1 volt
B. 10 volts
C. 11 volts
D. 9 volts

T5D12. What is the voltage across a 10-ohm resistor if a current of 2 amperes flows through it?
A. 8 volts
B. 0.2 volts
C. 12 volts
D. 20 volts

T5D13. In which type of circuit is the current always the same through all components?
A. Series
B. Parallel
C. Resonant
D. Branch

T5D14. In which type of circuit is the voltage always the same across all components?
A. Series
B. Parallel
C. Resonant
D. Branch

Answer Key

T5B01: C	T5B02: A	T5B03: C	T5B04: A	T5B05: B
T5B06: D	T5B07: C	T5B08: B	T5B09: B	T5B10: C
T5B11: A	T5B12: D	T5B13: C	T5C01: D	T5C02: A
T5C03: D	T5C04: C	T5C05: D	T5C06: A	T5C07: D
T5C08: A	T5C09: A	T5C10: B	T5C11: B	T5C12: A
T5D01: B	T5D02: A	T5D03: B	T5D04: B	T5D05: C
T5D06: A	T5D07: D	T5D08: C	T5D09: C	T5D10: A
T5D11: B	T5D12: D	T5D13: A	T5D14: B	